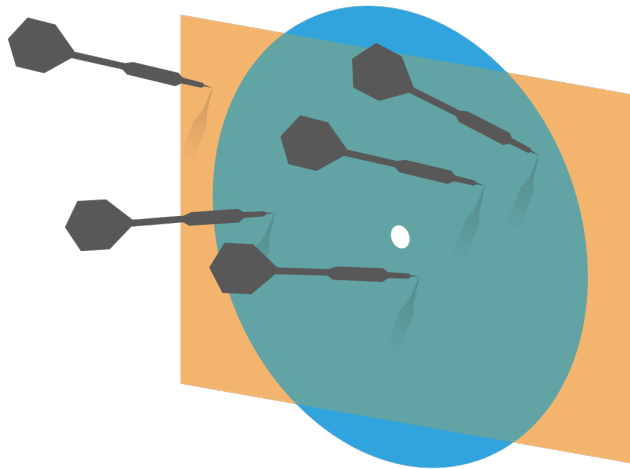


Probability and Statistics for Computer Science



“The weak law of large numbers gives us a very valuable way of thinking about expectations.” ---Prof. Forsythe

Credit: wikipedia

Last time

✱ Random Variable

- ✱ Conditional probability and joint probability

- ✱ *Independence of random variables*

- ✱ *Expected value*

Objectives

* Random Variable

- * Review

- * Variance and Covariance

- * *The weak law of large numbers*

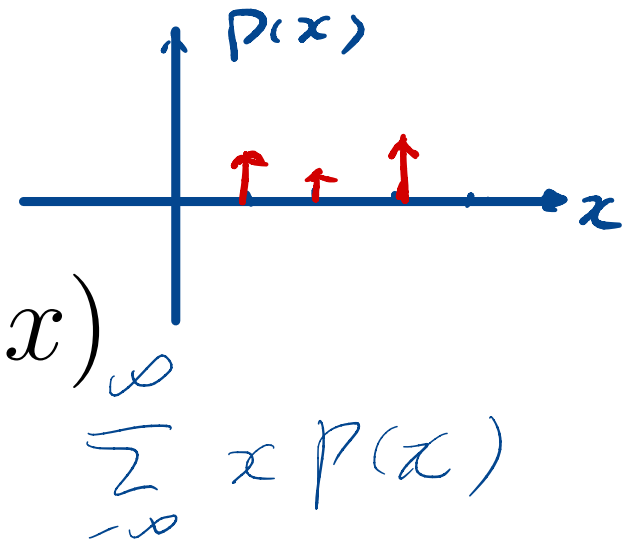
- * *Simulation & example of airline overbooking*

Expected value

- ✱ The **expected value** (or **expectation**) of a random variable X is

$$E[X] = \sum_{-\infty}^{\infty} x P(x)$$

Theoretical mean



The expected value is a **weighted sum** of **all** the values X can take

Linearity of Expectation

$$E[ax + bY] = a E[X] + b E[Y]$$

$$E\left[\sum_i c_i X_i\right] = \sum_i c_i E[X_i]$$

Expected value of a function of X

$$E[f(X)] = \sum_x f(x) p(x)$$

$$E[f(X, Y)] = \sum_x \sum_y f(x, y) \underbrace{p(x, y)}_{p(X=x \cap Y=y)}$$

Variance and standard deviation

✱ The variance of a random variable X is

$$f(x) = (x - E[X])^2$$

$$\text{var}[X] = E[(X - E[X])^2]$$

✱ The standard deviation of a random variable X is

$$\text{std}[X] = \sqrt{\text{var}[X]} = \sum_x f(x) p(x) = \sum_x (x - E[X])^2 p(x)$$

Properties of variance

✱ For random variable X and constant k

$$\text{var}[X] \geq 0$$

$$\text{var}[kX] = k^2 \text{var}[X]$$

If a RV X has a variance of 4.

$$Y = 3X$$

what's $\text{var}[Y]$?

if Z has the same distribution
as Y , what's $\text{var}[Z]$?

A neater expression for variance

- ✱ Variance of Random Variable X is defined as:

$$\begin{aligned} \text{var}[X] &= E[(X - E[X])^2] \\ &= \sum_x (x - E[X])^2 \cdot p(x) \end{aligned}$$

- ✱ It's the same as:

$$\text{var}[X] = E[X^2] - E[X]^2$$

A neater expression for variance

$$\textit{var}[X] = E[(X - E[X])^2]$$

Neat expression of $\text{Var}[X]$

$$\text{var}[X] = E[(X - E[X])^2]$$

$$= E[X^2 - 2XE[X] + (E[X])^2]$$

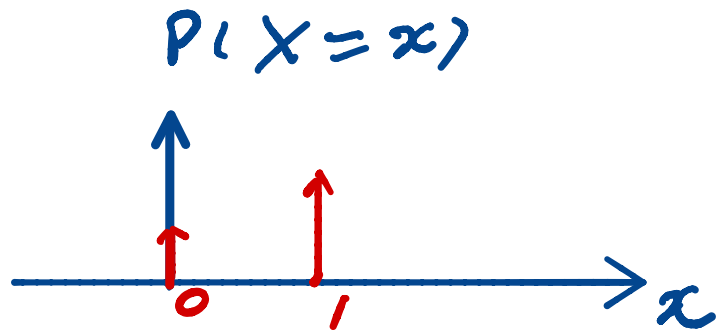
$$= E[X^2] - 2E[X] \cdot E[X] + (E[X])^2$$

$$= E[X^2] - 2(E[X])^2 + (E[X])^2$$

$$= E[X^2] - (E[X])^2$$

$$= \sum_x x^2 p(x) - \left(\sum_x x p(x) \right)^2$$

What are the $E[X]$ and $\text{Var}[X]$?



$$P(X=x) = \begin{cases} a & x=0 \\ 2a & x=1 \\ 0 & \text{ow} \end{cases}$$

$$a = ?$$

$$a + 2a = 1$$

due to

$$P(\Omega) = 1$$

$$\sum_x P(x) = 1$$

$$E[X] = ?$$

$$\sum x P(x) \\ 0 \times \frac{1}{3} + 1 \times \frac{2}{3} = \frac{2}{3}$$

$$\text{Var}[X] = ?$$

$$\begin{aligned} & E[X^2] - (E[X])^2 \\ &= 0^2 \times \frac{1}{3} + 1^2 \times \frac{2}{3} - \left(\frac{2}{3}\right)^2 = \frac{2}{9} \end{aligned}$$

Motivation for covariance

- ✱ Study the relationship between random variables
- ✱ Note that it's the un-normalized correlation
- ✱ Applications include the fire control of radar, communicating in the presence of noise.

Covariance

✱ The **covariance** of random

variables X and Y is $\hat{X} = \frac{x - E[X]}{\sigma_x}$

$$\text{cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

$\sum_x \sum_y f(x, y) p(x, y)$

✱ Note that

$$\text{cov}(X, X) = E[(X - E[X])^2] = \text{var}[X]$$

$\frac{1}{N} \sum \hat{x} \hat{y}$

A neater form for covariance

- ✱ A neater expression for **covariance** (similar derivation as for variance)

$$\begin{aligned} \text{cov}(X, Y) &= \underline{E[XY] - E[X]E[Y]} \\ &= \sum_x \sum_y xy p(x, y) - E[X] \cdot E[Y] \end{aligned}$$

Correlation coefficient is normalized covariance

✱ The correlation coefficient is

$$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

✱ When X, Y takes on values with equal probability to generate data sets $\{(x, y)\}$, the correlation coefficient will be as seen in Chapter 2.

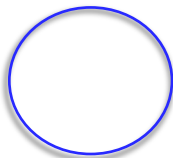
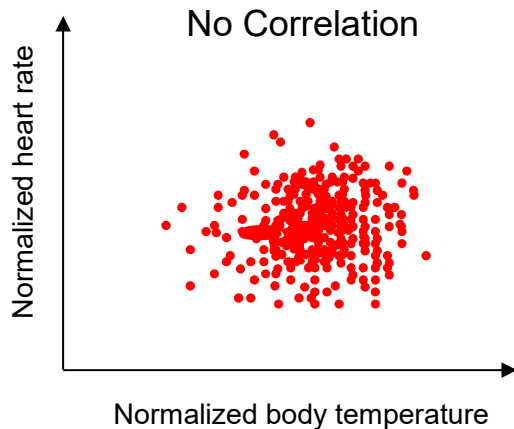
Correlation coefficient is normalized covariance

- ✱ The correlation coefficient can also be written as:

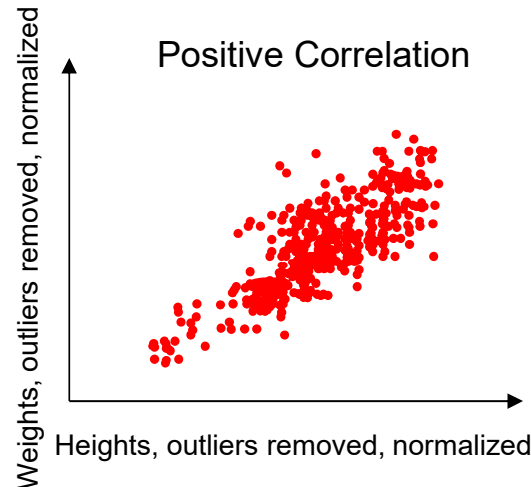
$$\text{corr}(X, Y) = \frac{E[XY] - E[X]E[Y]}{\sigma_X \sigma_Y}$$

Correlation seen from scatter plots

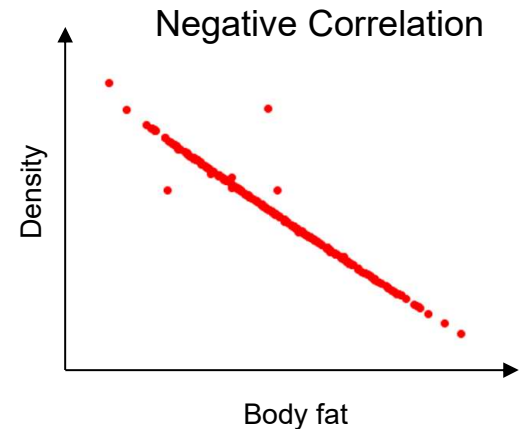
Zero Correlation



Positive correlation

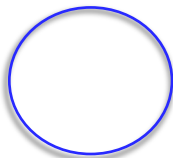
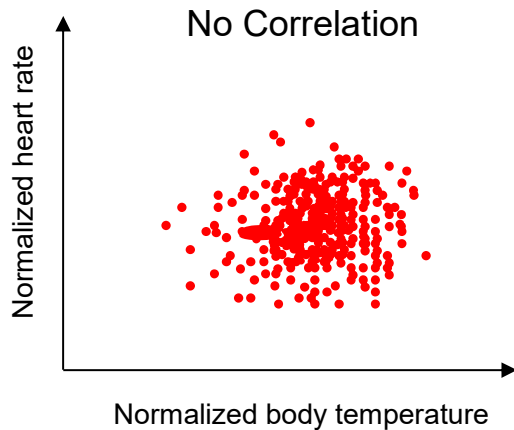


Negative correlation

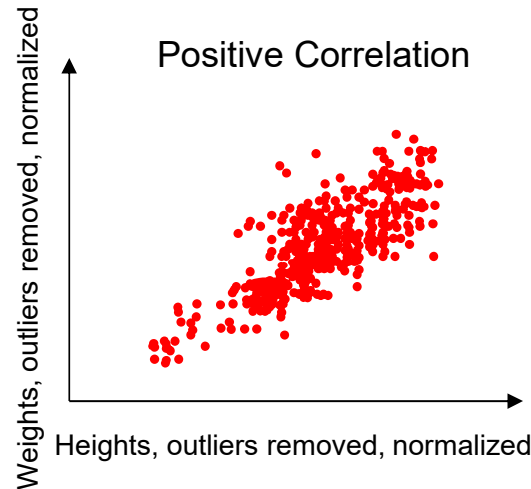


Covariance seen from scatter plots

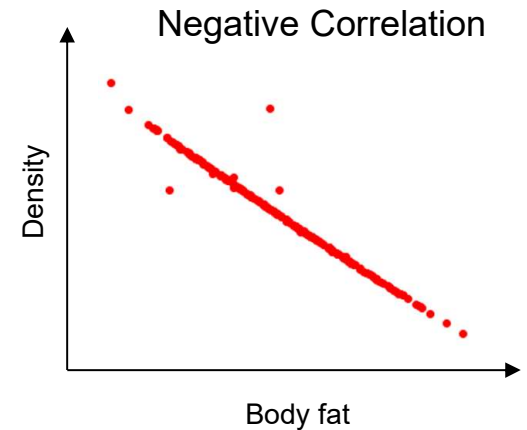
Zero Covariance



Positive Covariance



Negative Covariance



When correlation coefficient or covariance is zero

- ✱ The covariance is 0!

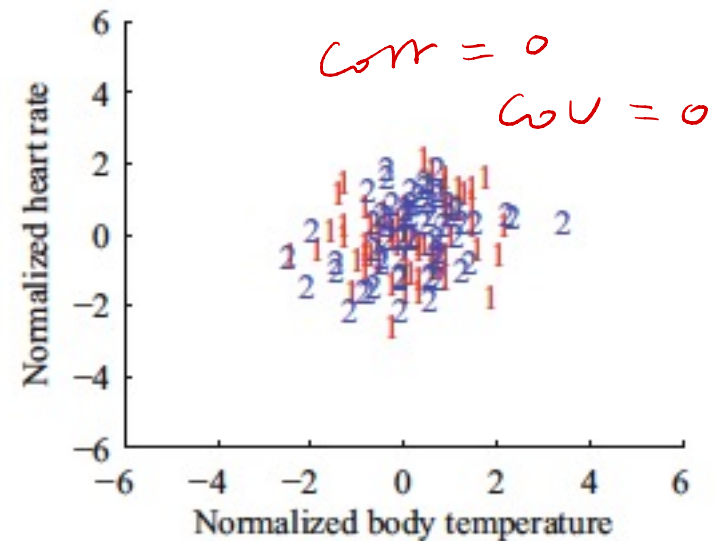
- ✱ That is:

$$E[XY] - E[X]E[Y] = 0$$

$$E[XY] = E[X]E[Y]$$

- ✱ This is a necessary property of independence of random variables * (not equal to independence) Not sufficient for indpt.!

Uncorrelatedness
No Correlation



Variance of the sum of two random variables

$$\text{var}[X + Y] = \text{var}[X] + \text{var}[Y] + 2\text{cov}(X, Y)$$

$$\text{var}[X + Y] = E[(X + Y)^2] - (E[X + Y])^2$$

If RVs X & Y are independent, then

✱ $E[XY] = E[X]E[Y]$ If X, Y indpt

$\downarrow$

$$\begin{aligned} \text{LHS} &= \sum_x \sum_y xy P(x, y) \\ &= \sum_x \sum_y xy \cdot P(x) P(y) \\ &= \sum_x x P(x) \sum_y y P(y) \\ &= E[X] E[Y] \end{aligned}$$

$\downarrow$

$$\begin{aligned} P(X, Y) \\ &= P(x) P(y) \end{aligned}$$

If X, Y indep $\Rightarrow E[XY] = E[X]E[Y]$ ^①

$$\therefore \text{cov}(X, Y) = E[XY] - E[X]E[Y]$$

due to ①

$$= E[XY] - E[X]E[Y]$$

X, Y

$$= 0 \Rightarrow \text{uncorrelated!}$$

then

$$\text{var}[X+Y] = \text{var}[X] + \text{var}[Y] + 2\cancel{\text{cov}(X, Y)}$$
$$= \text{var}[X] + \text{var}[Y]$$

These are equivalent!

Uncorrelatedness

✱ $E[XY] = E[X]E[Y]$ if X, Y are uncorrelated.

$cov(X, Y) = 0$ if X, Y are uncorrelated.
 $corr(X, Y) = 0$

$var[X + Y] = var[X] + var[Y]$ if X, Y are uncorrelated.

Q: What is this expectation?

- ✱ We toss two fair identical coins A & B independently for three times and 4 times respectively, for each head we earn \$1, we define X is the earning from A and Y is the earning from B. What is $E[XY]$?

A. 2

B. 3

C. 4

$$\begin{aligned} &= E[X] E[Y] \\ &= \left(\frac{1}{2} \times 3\right) \cdot \left(\frac{1}{2} \times 4\right) \\ &= 3 \end{aligned}$$

X_1 : carrying from
1st toss
of A

$$X = X_1 + X_2 + X_3$$

$$E[X] = E[X_1] + E[X_2] + E[X_3]$$

$$= \frac{1}{2} \times 3$$

$$= \frac{3}{2}$$

$$\begin{aligned} E[X_1] &= 1 \times p + 0 \times (1-p) \\ &= p = 0.5 \end{aligned}$$

$$E[X_2] = E[X_1]$$

$$E[X_3] = E[X_1]$$

Try at home

Uncorrelated vs Independent

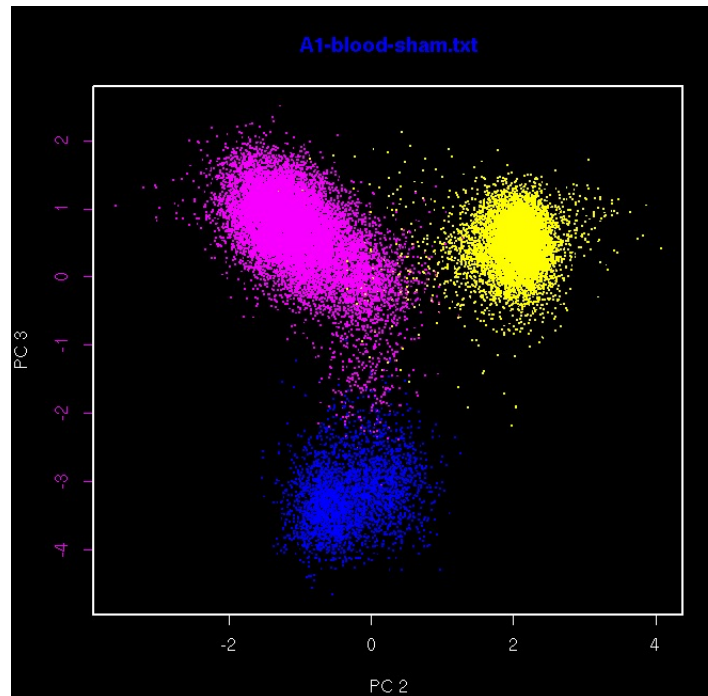
✱ If two random variables are uncorrelated, does this mean they are independent? Investigate the case X takes $-1, 0, 1$ with equal probability and $Y=X^2$.

✓ Are X, Y indpt?

② Are x, Y uncorrelated?

Covariance example

It's an underlying concept in principal component analysis in Chapter 10



Towards the weak law of large numbers

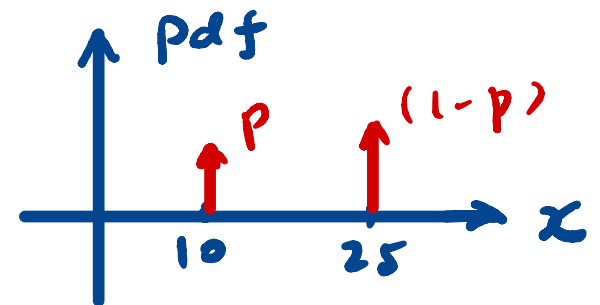


Random Variable Example for WLLN



Money box

10¢ 25¢
dime quarter



* shake and take one
and put back
(with replacement)

$$X_1 = 10? 25?$$

$$X_2 = 10? 25?$$

$$\vdots$$
$$X_n$$

$$E[X] = ?$$

$$10p + 25(1-p)$$

$$\bar{X}_n = \frac{\sum_{i=1}^n X_i}{n}$$

Convergence in Probability

$$X(n) \rightarrow U$$

$$\lim_{n \rightarrow \infty} P(|X(n) - U| > \varepsilon) \rightarrow 0$$

$$n \rightarrow \infty \quad \lim_{n \rightarrow \infty} P(|X(n) - U| < \varepsilon) \rightarrow 1 \quad (\forall \varepsilon > 0)$$

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - E[X]| > \varepsilon) \rightarrow 0$$

Markov's inequality

- ✱ For any random variable X that *only* takes $x \geq 0$ and constant $a > 0$

$$P(X \geq a) \leq \frac{E[X]}{a}$$

- ✱ For example, if $a = 10 E[X]$

$$P(X \geq 10E[X]) \leq \frac{E[X]}{10E[X]} = 0.1$$

Proof of Markov's inequality



$$E[X] = \sum_x x p(x) = \sum_{x \in (-\infty, 0)} x p(x) + \sum_{x \in [0, a)} x p(x) + \sum_{x \in [a, \infty)} x p(x)$$

$$\Rightarrow E[X] \geq \sum_{x \in [a, \infty)} x p(x) \geq \sum_{x \in [a, \infty)} a p(x)$$

$$E[X] \geq a \cdot p(X \geq a)$$

$$\therefore p(X \geq a) \leq \frac{E[X]}{a}$$

$$= a \sum_{a \leq x < \infty} p(x)$$

$$= a p(X \geq a)$$

Chebyshev's inequality

- ✱ For **any** random variable X and constant $a > 0$

$$P(|X - E[X]| \geq a) \leq \frac{\text{var}[X]}{a^2}$$

- ✱ If we let $a = k\sigma$ where $\sigma = \text{std}[X]$

$k > 0$

$$P(|X - E[X]| \geq k\sigma) \leq \frac{1}{k^2}$$

- ✱ In words, the probability that X is greater than k standard deviation away from the mean is small

Proof of Chebyshev's inequality

✱ Given Markov inequality, $a > 0$, $x^* \geq 0$

$$P(X^* \geq a) \leq \frac{E[X^*]}{a}$$

let $|U| = X^*$ $w = a$

✱ We can rewrite it as

$$w > 0 \quad P(|U| \geq w) \leq \frac{E[|U|]}{w}$$

for any U

Proof of Chebyshev's inequality

✱ If $U = (X - E[X])^2$ X is any RV.

$$P(|U| \geq w) \leq \frac{E[|U|]}{w} = \frac{E[U]}{w}$$

$$P((X - E[X])^2 \geq w) \leq \frac{E[(X - E[X])^2]}{w}$$

$$\text{LHS} = P(|X - E[X]| \geq a) = \frac{\text{Var}[X]}{w} = \frac{\text{Var}[X]}{a^2}$$

$$w = a^2, a > 0$$

Proof of Chebyshev's inequality

✱ Apply Markov inequality to $U = (X - E[X])^2$

$$P(|U| \geq w) \leq \frac{E[|U|]}{w} = \frac{E[U]}{w} = \frac{\text{var}[X]}{w}$$

X is any R.V.

✱ Substitute $U = (X - E[X])^2$ and $w = a^2$

$$P((X - E[X])^2 \geq a^2) \leq \frac{\text{var}[X]}{a^2} \quad \text{Assume } a > 0$$

$$\Rightarrow P(|X - E[X]| \geq a) \leq \frac{\text{var}[X]}{a^2}$$

Now we are closer to the law of large numbers



Sample mean and IID samples

- ✱ We define the sample mean $\bar{X}$ to be the average of N random variables $X_1, \dots, X_N$.

- ✱ If $X_1, \dots, X_N$ are ***independent*** and have ***identical*** probability function $P(x)$

then the numbers randomly generated from them are called **IID** samples

- ✱ The **sample mean** is a random variable

$$\bar{X} = \frac{\sum X_i}{N}$$

Sample mean and IID samples

- ✱ Assume we have a set of **IID samples** from **N** random variables $X_1, \dots, X_N$ that have probability function $P(x)$
- ✱ We use $\overline{\mathbf{X}}$ to denote the **sample mean** of these **IID samples**

$$\overline{\mathbf{X}} = \frac{\sum_{i=1}^N X_i}{N}$$

Expected value of sample mean of IID random variables

✱ By linearity of expected value

$$E[\bar{X}] = E\left[\frac{\sum_{i=1}^N X_i}{N}\right] = \frac{1}{N} \sum_{i=1}^N E[X_i]$$

X_i they have identical pdf

$$E[X_i] = E[X]$$

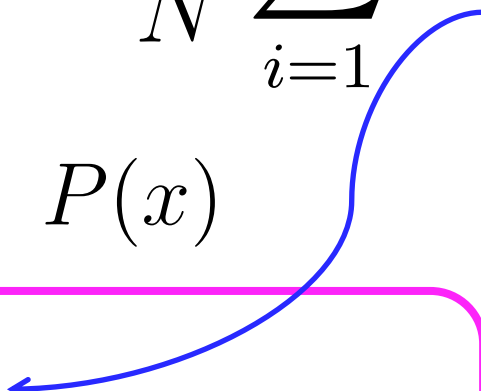
$$= \frac{1}{N} \cdot N \cdot E[X] = E[X]$$

Expected value of sample mean of IID random variables

✱ By linearity of expected value

$$E[\bar{\mathbf{X}}] = E\left[\frac{\sum_{i=1}^N X_i}{N}\right] = \frac{1}{N} \sum_{i=1}^N E[X_i]$$

✱ Given each X_i has identical $P(x)$


$$E[\bar{\mathbf{X}}] = \frac{1}{N} \sum_{i=1}^N E[X] = E[X]$$

Variance of sample mean of IID random variables

✱ By the scaling property of variance

$$\text{var}[\bar{\mathbf{X}}] = \text{var}\left[\frac{1}{N} \sum_{i=1}^N X_i\right] = \left(\frac{1}{N^2}\right) \text{var}\left[\sum_{i=1}^N X_i\right]$$

if X, Y indpt

$$\text{var}[X+Y] = \text{var}[X] + \text{var}[Y]$$

if " X_i " mutually indpt

$$\text{var}\left[\sum X_i\right] = \sum \text{var}[X_i]$$

$$\text{var}[X_i] = \text{var}[X]$$

$X_i, \text{ iid}$

$$\begin{aligned} &= \frac{1}{N^2} \sum_{i=1}^N \text{var}[X_i] \\ &= \frac{1}{N^2} \cdot N \cdot \text{var}[X] \\ &= \frac{1}{N} \cdot \text{var}[X] \end{aligned}$$

Variance of sample mean of IID random variables

✱ By the scaling property of variance

$$\text{var}[\bar{\mathbf{X}}] = \text{var}\left[\frac{1}{N} \sum_{i=1}^N X_i\right] = \left(\frac{1}{N^2}\right) \text{var}\left[\sum_{i=1}^N X_i\right]$$

✱ And by independence of these IID random variables

$$\text{var}[\bar{\mathbf{X}}] = \frac{1}{N^2} \sum_{i=1}^N \text{var}[X_i]$$

Variance of sample mean of IID random variables

✱ By the scaling property of variance

$$\text{var}[\bar{\mathbf{X}}] = \text{var}\left[\frac{1}{N} \sum_{i=1}^N X_i\right] = \left(\frac{1}{N^2}\right) \text{var}\left[\sum_{i=1}^N X_i\right]$$

✱ And by independence of these IID random variables

$$\text{var}[\bar{\mathbf{X}}] = \frac{1}{N^2} \sum_{i=1}^N \text{var}[X_i]$$

✱ Given each X_i has identical $P(x)$, $\text{var}[X_i] = \text{var}[X]$

$$\text{var}[\bar{\mathbf{X}}] = \frac{1}{N^2} \sum_{i=1}^N \text{var}[X] = \frac{\text{var}[X]}{N}$$

Expected value and variance of sample mean of IID random variables

- ✱ The expected value of sample mean is the same as the expected value of the distribution

$$E[\bar{X}] = E[X]$$

- ✱ The variance of sample mean is the distribution's variance divided by the sample size **N**

$$\text{var}[\bar{X}] = \frac{\text{var}[X]}{N}$$

Weak law of large numbers

✱ Given a random variable X with finite variance, probability distribution function $P(x)$ and the sample mean $\bar{X}$ of size N .

✱ For any positive number $\epsilon > 0$

$$\lim_{N \rightarrow \infty} P(|\bar{X} - E[X]| \geq \epsilon) = 0$$

✱ That is: the value of the mean of **IID** samples is very close with high probability to the expected value of the population when sample size is very large

Proof of Weak law of large numbers

✱ Apply Chebyshev's inequality

$$P(|\bar{\mathbf{X}} - \underbrace{E[\bar{\mathbf{X}}]}_{E[X]}| \geq \epsilon) \leq \frac{\text{var}[\bar{\mathbf{X}}]}{\epsilon^2}$$

$\epsilon > 0$

$$\text{RHS} = \frac{\text{var}[X]}{N \epsilon^2}$$

lim $N \rightarrow \infty$

$$P(|\bar{x} - E[X]| \geq \epsilon) \leq \frac{\text{var}[X]}{N \epsilon^2} \rightarrow 0$$

lim $N \rightarrow \infty$

$$P(|\bar{x} - E[X]| < \epsilon) = 1$$

$\forall \epsilon > 0$

Proof of Weak law of large numbers

- ✱ Apply Chebyshev's inequality

$$P(|\bar{\mathbf{X}} - E[\bar{\mathbf{X}}]| \geq \epsilon) \leq \frac{\text{var}[\bar{\mathbf{X}}]}{\epsilon^2}$$

- ✱ Substitute $E[\bar{\mathbf{X}}] = E[X]$ and $\text{var}[\bar{\mathbf{X}}] = \frac{\text{var}[X]}{N}$

Proof of Weak law of large numbers

✱ Apply Chebyshev's inequality

$$P(|\bar{\mathbf{X}} - E[\bar{\mathbf{X}}]| \geq \epsilon) \leq \frac{\text{var}[\bar{\mathbf{X}}]}{\epsilon^2}$$

✱ Substitute $E[\bar{\mathbf{X}}] = E[X]$ and $\text{var}[\bar{\mathbf{X}}] = \frac{\text{var}[X]}{N}$

$$P(|\bar{\mathbf{X}} - E[\mathbf{X}]| \geq \epsilon) \leq \frac{\text{var}[\mathbf{X}]}{N\epsilon^2}$$

Proof of Weak law of large numbers

✱ Apply Chebyshev's inequality

$$P(|\bar{\mathbf{X}} - E[\bar{\mathbf{X}}]| \geq \epsilon) \leq \frac{\text{var}[\bar{\mathbf{X}}]}{\epsilon^2}$$

✱ Substitute $E[\bar{\mathbf{X}}] = E[X]$ and $\text{var}[\bar{\mathbf{X}}] = \frac{\text{var}[X]}{N}$

$$P(|\bar{\mathbf{X}} - E[\mathbf{X}]| \geq \epsilon) \leq \frac{\text{var}[\mathbf{X}]}{N\epsilon^2} \xrightarrow[N \rightarrow \infty]{} 0$$

Proof of Weak law of large numbers

✱ Apply Chebyshev's inequality

$$P(|\bar{\mathbf{X}} - E[\bar{\mathbf{X}}]| \geq \epsilon) \leq \frac{\text{var}[\bar{\mathbf{X}}]}{\epsilon^2}$$

✱ Substitute $E[\bar{\mathbf{X}}] = E[X]$ and $\text{var}[\bar{\mathbf{X}}] = \frac{\text{var}[X]}{N}$

$$P(|\bar{\mathbf{X}} - E[X]| \geq \epsilon) \leq \frac{\text{var}[X]}{N\epsilon^2} \xrightarrow[N \rightarrow \infty]{} 0$$

$$\lim_{N \rightarrow \infty} P(|\bar{\mathbf{X}} - E[X]| \geq \epsilon) = 0$$

Assignments

- ✱ Continue to work on HW4
- ✱ Module Week 5
- ✱ Next time: Continuous random variable, classic known probability distributions

Additional References

- ✱ Charles M. Grinstead and J. Laurie Snell
"Introduction to Probability"
- ✱ Morris H. Degroot and Mark J. Schervish
"Probability and Statistics"

See you next time

*See
You!*

